Ashish Lukka

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Education

VNR Vignana Jyothi Institute of Engineering & Technology — Hyderabad
Bachelor of Technology in Information Technology — *CGPA: 9.15* — *2023 – 2027*

Technical Skills

Programming Languages: Java, JavaScript, Python, C
Web Development: React.js, HTML, CSS, Bootstrap, Node.js, Express.js, REST APIs
Databases: MongoDB, MySQL, Neo4j
Tools & Methodologies: Git, GitHub, Postman, Notion, Agile/Scrum
Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks, System Design, Machine Learning

Experience

Full Stack Developer Intern — June 2025 – July 2025
Olive Crypto Systems — *Hyderabad*

- Built course management with nested hierarchy (Course → Units → Assessments) and approval-based workflow
- Designed RESTful APIs and MongoDB schemas with embedded progress tracking, paginated course listing
- Implemented role-based authentication and authorization with Microsoft OAuth/SSO for secure access control
- Developed secure assessment system with randomized sets, violation-based blocking, and admin control
- Built automated certification engine with Canvas-based generation, QR-code verification, public validation page.

Projects

Sales Ticket | *Live Demo* | *GitHub*

- Built a full-stack platform that converts **customer support tickets** into **actionable business signals** such as **churn risk** and **upsell opportunities**
- Designed a **ticket processing pipeline** that fetches, classifies, and routes each ticket, then drafts a reply using **Groq** LLMs and sends it to the right team persona
- Used **MySQL** for ticket data and **Neo4j** for account relationships and health tracking, served through **20+ REST APIs**
- Built a **RAG-based assistant** that finds similar past tickets using **embeddings and cosine similarity**, and keeps session memory for follow-up questions
- Developed a **7-view React dashboard** with **role-based access** to visualize tickets, accounts, and signals
- **Tech Stack:** React, Node.js, Express.js, MySQL, Neo4j, Groq API, Xenova Transformers, JWT, Zendesk API

Parameter-Efficient Domain Adaptation for Speech Recognition

- Developed a fully open-source ASR domain adaptation framework that fine-tunes **OpenAI Whisper** using **LoRA** adapters trained exclusively on synthetically generated speech data across **music, weather, and sports** domains
- Built an automated **Llama3-8B + Tacotron2-DDC** pipeline to generate over **121,000 synthetic audio-transcript pairs** (114+ hours of speech), eliminating reliance on manually annotated domain-specific datasets
- Achieved up to **29.3% relative WER reduction** and outperformed Meta AI's DAS framework on 2 of 3 domains, evaluated on **5,640 real-world speech samples** from Mozilla Common Voice 17.0
- **Tech Stack:** Python, Whisper, LoRA, PEFT, Llama3-8B, Tacotron2-DDC

Draft - Blog App | *Live Demo* | *GitHub*

- Architected a scalable blog platform with 3-role RBAC (Admin, Author, Reader) and role-specific dashboards
- Developed 10+ REST API endpoints across admin, author, and user modules to support article publishing, editing, deletion, commenting, user verification, and account status management.
- **Tech Stack:** React.js, Node.js, Express.js, MongoDB, REST APIs, Clerk, Bootstrap, Git

Certifications & Achievements

Certifications: AWS Cloud Quest: Cloud Practitioner, Cloud Computing – NPTEL, Postman API Fundamentals Student Expert
Achievements: Competed in 6+ hackathons, secured 1st place at Webathon 3.0 (ACM VNRVJIET), and 2nd place at Web Alchemy (CSI VNRVJIET)